Dear Editor,

We are writing to submit our manuscript entitled “Seasonal influenza viruses show distinct adaptive dynamics during growth in chicken eggs”, which we hope you will consider for publication in Molecular Biology and Evolution.

Resolving the fitness landscape on which viruses evolve is crucial both for designing vaccines and antivirals and for understanding the fundamental biological force of adaptation. In antigenically evolving pathogenic viruses, such as influenza, the fitness landscape has aptly referred to as a seascape, referring to how fitness peaks change as evolution traverses genetic space and also as selective immune pressures vary. We know a lot about how human influenza virus has evolved from the genetic sequences of the virus collected over the past decades; however, we know less about what the fitness landscape looked like over time. In other words, we can see what did happen, but not what could have happened. This leaves open the question of whether, at any given time, influenza has hundreds of adaptive trajectories available to it or only a couple.

In this manuscript, we aim to address this question of how epistatic constraints and stochasticity dictate evolution by looking at repeated adaptation of human influenza virus to chicken eggs. The majority of influenza vaccine doses are produced by growing the virus in chicken eggs, and this process is known to result in some egg-adaptive mutations. We have thousands of sequences of human influenza that were grown in eggs over the past several decades. Some of the egg-passaged strains in this dataset originate from human isolates with identical sequences, while others derive from viruses that are genetically different due to rapid antigenic evolution during this timeframe. This allows us to consider the effects of stochasticity and epistasis on egg-adaptation, respectively.

We find that, throughout the influenza genome, almost all egg-adaptive mutations occur within hemagglutinin (HA) and that these mutations are concentrated at the receptor-binding pocket of this protein, suggesting that differences in the human and egg sialic acid receptors are a strong selective pressure for egg-passaged influenza. We also observe that egg-adaptive mutations change over time and correlate with fixations in the human influenza HA. We see epistatic effects not only between egg-adaptive mutations and the genetic background, but also between egg-adaptive mutations. This epistasis constrains egg-adaptive evolution such that multiple, mutually exclusive, fitness peaks exist at any given time. We find that replicates of identical HA sequences will evolve to different fitness peaks, supporting the role of stochasticity in selecting an adaptive trajectory.

We observe consistent differences in egg-adaptation of the four human influenza subtypes, with A/H3N2 exhibiting the most background dependence, the most positively-epistatic interactions, and the longest adaptive walks, and the influenza B viruses (B/Vic and B/Yam) having the least of all of these with A/H1N1pdm somewhere in between. This ordering mirrors the relative rates of antigenic drift of these subtypes, suggesting that human influenza A viruses (and particularly H3N2) have more adaptable HA proteins, allowing them to more readily access genetic changes that improve binding to egg sialic receptors and also to evade antibody recognition while maintaining high receptor-binding capabilities.

Thank you for considering our manuscript for publication in Molecular Biology and Evolution.

We look forward to your response. Sincerely,

Kathryn Kistler and Trevor Bedford